the term Λg_{ab} in the equation written above. But it is a further mistake: the added term is correct, but it does not change the fact that the equation predicts that the universe must be expanding. For all his bravery, Einstein the genius lacks the courage to believe his own equations.

A few years later Einstein is forced to give up: it is his theory that is right, not his reservations about it. Astronomers realize that all galaxies are indeed moving away from us. The universe is expanding, exactly as the equations predicted. Fourteen billion years ago, the universe was concentrated almost to a single, furiously hot point. From there it expanded in a colossal ‘cosmic’ explosion – and here the term ‘cosmic’ is not used in any rhetorical sense: it is, literally, a cosmic explosion. This is the ‘Big Bang’.

Today we know the expansion is real. The definitive proof of the scenario foreseen by Einstein’s equations arrives in 1964, when two American radio-astronomers, Arno Penzias and Robert Wilson, discover by accident a radiation diffused throughout the universe which turns out to be precisely what remains of the original immense heat of the early universe. Once again, the theory turns out to have been correct, up to its most amazing predictions.

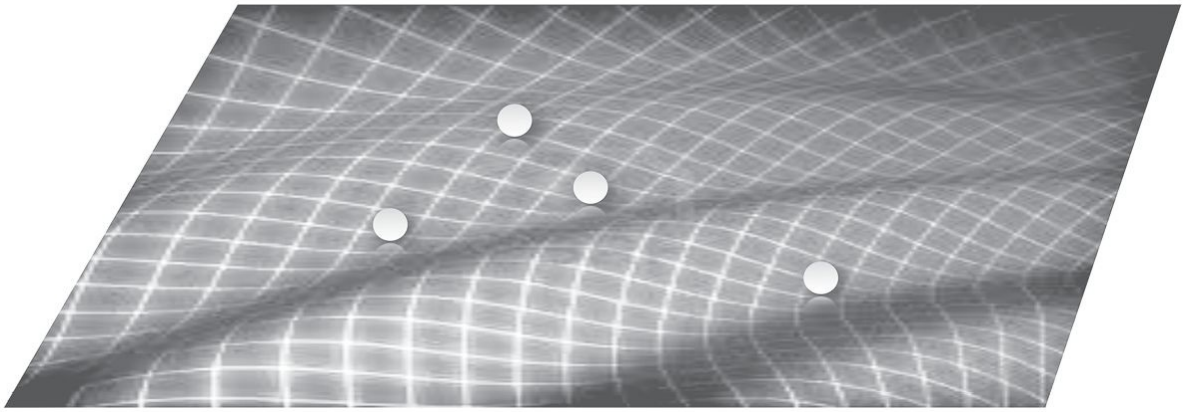


Figure 3.17 Einstein’s world: particles and fields which move on other fields.

Ever since we discovered that the Earth is round and turns like a mad spinning-top, we have understood that reality is not what it seems: every time we glimpse a new aspect of it, it is a deeply emotional experience. Another veil has fallen. But the leap made by Einstein is unparalleled: spacetime is a field; the world is made only of fields and particles; space and time are not something else, something different from the rest of nature: they are just a field among the others (figure 3.17).

In 1953, a primary schoolchild writes to Albert Einstein, ‘Our class is studying the universe. I am very interested in space. I would like to thank you for all that you have done so that we might understand it.’[5]

I feel the same way.